

Highlights

FOLoRA: Fisher-Orthogonalized Low-Rank Adaptation for Continual Learning

- FOLoRA protects the Fisher-important directions of a shared LoRA adapter.
- Protected directions are weighted by Fisher importance, not protected equally.
- FOLoRA minimizes the leading term of a second-order bound with explicit remainder.
- Fixed trainable-parameter budget with no replay and no inference overhead.

FOLoRA: Fisher-Orthogonalized Low-Rank Adaptation for Continual Learning

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Abstract

Parameter-efficient fine-tuning (PEFT) methods such as low-rank adaptation (LoRA) freeze a pre-trained backbone and train few parameters, which suits continual learning on resource-constrained devices. However, sequentially fine-tuning a shared LoRA adapter still forgets catastrophically: the low-rank subspace is overwritten by new tasks. We propose **FOLoRA** (Fisher-Orthogonalized Low-Rank Adaptation), a replay-free, fixed-budget method protecting the directions past tasks deem important. FOLoRA maintains, per adapted layer, an accumulated second-order importance kernel from the empirical Fisher restricted to the adapter parameters. Its top eigendirections form an *importance-weighted protected subspace*; during training a regularizer penalizes alignment of the adapter displacement with them, weighted by eigenvalue. This yields a stability–plasticity trade-off: important directions are preserved, unimportant ones stay reusable. We further derive a second-order forgetting bound with an explicit higher-order remainder, and show FOLoRA’s regularizer minimizes its leading term. We audit that bound’s slack empirically, using it as a design principle for *what* to penalize rather

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than a predictor of *how much* it helps. FOLoRA matches the strongest regularized baseline on Split CIFAR-100 and Split ImageNet-R under the frozen-feature nearest-class-mean protocol. Unlike per-task-adapter methods (O-LoRA, InfLoRA), FOLoRA needs neither replay nor parameter growth with task count ($\sim 18\times$ fewer trainable parameters) and adds no inference cost, as the update merges into the backbone. We also characterize the trade-off: a rank- k importance model rather than EWC’s full diagonal costs a larger footprint, which we quantify alongside accuracy.

Keywords: continual learning, catastrophic forgetting, parameter-efficient fine-tuning, low-rank adaptation, Fisher information

1. Introduction

Continual learning aims to learn a sequence of tasks without catastrophically forgetting previously acquired knowledge [1]. The dominant strategies — rehearsal [2], regularization [3, 4], and dynamic architectures [5] — were largely developed in the regime of training a model from scratch. Recently, however, the community has shifted toward *pre-trained models* (e.g., Vision Transformers) combined with *parameter-efficient fine-tuning* (PEFT), in which the backbone is frozen and only a tiny set of parameters is learned per task [6, 7, 8]. This shift has produced a fast-moving literature on prompt-, adapter-, and subspace-based PEFT for continual learning [9, 10, 11, 12, 13].

Low-rank adaptation (LoRA) [14] is a particularly attractive PEFT primitive for continual learning, and recent variants exploit it in different ways [15, 16, 17, 18]: a shared LoRA adapter keeps the total parameter count fixed regardless of the number of tasks, which is ideal for resource-constrained de-

ployment. However, sequentially fine-tuning a shared adapter still suffers from catastrophic forgetting — although the number of trainable parameters is small, the low-rank subspace is repeatedly overwritten, and the representational structure learned for earlier tasks is destroyed.

A line of recent work addresses this by allocating *orthogonal* low-rank subspaces to different tasks [8, 19], and refined task-wise allocation schemes continue to appear [20, 21, 22]. These methods prevent interference by making the new task’s adapter (nearly) orthogonal to those of previous tasks. While effective, they exhibit two limitations. First, they protect all directions of the past subspace *equally*, ignoring the fact that different directions carry vastly different amounts of information about a task. Second, most of them allocate a *new* adapter per task, so the parameter count grows linearly with the number of tasks.

In this paper we propose **FOLoRA** (Fisher-Orthogonalized Low-Rank Adaptation), a replay-free, fixed-budget method that mitigates forgetting by protecting past tasks’ *important* directions in proportion to their importance. Our contributions are:

1. We frame forgetting in low-rank continual learning through a second-order lens and derive an upper bound that decomposes forgetting into Fisher-weighted subspace overlaps. We audit the slack of that bound empirically rather than assuming it away, and use it as a design principle for what to penalize rather than as a numerical predictor of how much forgetting a given setting removes (Sec. 4).
2. We propose FOLoRA, which estimates a per-layer second-order importance kernel from the empirical Fisher information restricted to the

adapter parameters, forms an importance-weighted protected subspace from its top eigendirections, and regularizes the adapter to stay orthogonal to these directions (Sec. 3).

3. We show that FOLoRA’s regularizer is, up to a constant factor, exactly the Lagrangian penalty for constraining the leading term of a second-order forgetting bound, realizing a principled stability–plasticity trade-off, and that it contains equal-weight orthogonalization — the geometry underlying O-LoRA-style subspace methods — as the special case in which all protected directions carry equal weight.
4. On Split CIFAR-100 and Split ImageNet-R, FOLoRA matches the strongest regularized baseline in average accuracy with no replay buffer and no growth in parameters, while improving significantly over the parameter-efficient prompt and subspace baselines; we quantify the importance-model memory it pays for its rank- k subspace representation relative to EWC’s diagonal one, and we report the run-to-run reproducibility floor that sets the resolution of these comparisons (Sec. 5).

2. Related Work

Continual learning. Classical continual learning methods fall into three families. Rehearsal methods store and replay a subset of past data [2]; regularization methods add a penalty that slows down learning on parameters important to past tasks, most notably EWC via the diagonal Fisher information [3] and SI/MAS via synaptic importance [4, 23]; dynamic methods grow the network per task [5, 24]. Recent surveys of the field appear in [13, 12, 25], and the exemplar-free, representation-level, and distillation-based variants most

relevant here in [26, 27, 28, 29]. Our method is a regularization method, but it operates on the low-rank subspace of a frozen backbone rather than on raw parameters.

Parameter-efficient continual learning with pre-trained models. A recent and effective line of work keeps a pre-trained backbone frozen and learns only prompts or adapters; the space is surveyed in [9, 10, 11, 30]. Prompt-based methods — L2P [6], DualPrompt [31], and CODA-Prompt [7] — learn a pool of prompts and select or compose them per input. LoRA-based methods attach low-rank adapters. Related designs differ in where they spend the per-task budget: expandable or branched subspaces [20, 16, 17, 21], decoupled scaling rules [15], orthogonal adapter banks [32], further adapter variants [33, 34, 35, 36], dual-consolidation and hierarchical prompt decompositions [37, 38], and prototype- or prompt-routing schemes [39, 40, 41, 42]; for the online and task-free regime, see [18].

Orthogonalization and subspace allocation. A distinct line makes *where* a new task may write the design variable, rather than *how strongly*. Its classical form is gradient projection: orthogonal gradient descent [43], GPM [44] and the null-space-of-feature-covariance update of Adam-NSCL [45] retain a basis describing past tasks and constrain new updates to its orthogonal complement, a construction that also admits generalization guarantees [46]. The same idea reappears in PEFT as subspace allocation for low-rank adapters: O-LoRA [8] orthogonally initializes each task’s adapter against previous tasks’ input subspaces, and InfLoRA [19] places the new task’s subspace in the intersection of the new task’s gradient space and the orthogonal complement of the old tasks’ gradient space, removing interference while retaining

plasticity. Recent variants move the per-task budget along this axis — expandable or branched subspaces [20, 16, 17, 21] and null-space constructions for prompt tuning [47].

Two properties are shared across this line, and they are what FOLoRA changes. First, the protected subspace is obtained from a *structural* criterion — the past tasks’ input or feature covariance, a gradient subspace, or an explicit orthogonality constraint — rather than from the curvature of the past loss. Second, within the protected subspace every retained direction is defended equally: the budget decides *how many* directions to keep, never *how much* each one matters. FOLoRA keeps the subspace framing and replaces both choices with a single object, the adapter block of the accumulated empirical Fisher, so that the retained directions are those in which the past loss is most curved and each is penalized in proportion to its curvature. The equal-weight geometry of this family is then the special case $\sigma_j \equiv 1$ of FOLoRA’s penalty (Sec. 4), and we compare against O-LoRA and InfLoRA empirically in Sec. 5.2.

Second-order methods and curvature-weighted penalties. EWC [3] uses the diagonal Fisher as a per-parameter importance weight. FOLoRA combines that idea with the subspace framing above: it uses the top- k spectrum of the adapter block of the second-order expansion — not the diagonal Fisher, and not the full $d \times d$ matrix either — and applies it as a soft, importance-weighted orthogonal projection rather than a hard null-space constraint, which avoids the rank saturation that hard projections suffer once the null space shrinks. Recent analyses of flatness-aware optimization approach the same goal from other angles [48], flatness itself has been connected to gen-

eralization bounds [49], and gradient-norm regularization acts on the same stability–plasticity axis without a curvature weighting [50].

Theory of forgetting. Second-order analyses of forgetting are well established for models trained from scratch: the forgetting of a task is governed by the curvature of its loss landscape and by the overlap between successive tasks’ input distributions or NTKs [51, 52, 53, 54]. We build on this line but instantiate it in the low-rank adapter subspace of a frozen pre-trained model, and we use it constructively: the leading term of the bound is exactly the quantity that a subspace regularizer can minimize, which yields the FOLoRA objective (Sec. 4). Recent theory sharpens this picture for lazy-training and kernel regimes [55, 56], and the empirical counterpart — that low-rank adaptation forgets in a measurably different way from full fine-tuning [57] — is part of what motivates working in the adapter subspace rather than in the full parameter space.

Low-rank adaptation. LoRA [14] approximates weight updates by low-rank matrices and has become a standard PEFT primitive. Subsequent work studies initialization, rank allocation, and merging [36, 34]. We build directly on LoRA and extend it with a second-order anti-forgetting mechanism specialized for continual learning.

Evaluation protocols in PEFT-based continual learning. Results in this setting are strongly protocol-dependent. Keeping the standard trainable classification head that grows with the task sequence, class-incremental accuracy is typically in the single digits even when task-incremental accuracy is above 90%, because the loss is dominated by the classifier rather than by

the adapted features; replacing the head at evaluation time with a nearest-class-mean classifier over frozen features raises the same checkpoints into the 70–80% range. Two consequences matter for how a method should be tuned and compared. First, hyperparameters — the regularization strength in particular — are only identifiable relative to the protocol under which they were selected, and an optimum found under a classifier-dominated metric need not be the optimum under the reported one. Second, a ranking established under one protocol does not transfer to another. Accordingly, all defaults reported in this paper are re-selected under the frozen-feature nearest-class-mean protocol, and every comparison we draw is made within that protocol. That the protocol can dominate the reported conclusion is now being recognized independently: recent work shows the class-incremental average is sensitive to evaluation choices and can misrepresent representation quality [58, 59, 60], that classifier bias and calibration distort comparisons unless handled explicitly [61, 62], and that simple frozen-feature classifiers over a pre-trained backbone remain competitive with more elaborate designs [63, 64, 65].

Gap addressed in this work. None of the lines above supplies what this paper sets out to provide. Subspace-based methods [8, 19, 45] protect a past task’s subspace as a whole, so every retained direction is defended with the same strength regardless of how much the past loss actually depends on it, and most of them allocate a branch per task, so the trainable parameter count grows with the sequence length. Second-order analyses of forgetting [51, 54] characterize *why* forgetting happens but are developed for models trained from scratch and are not instantiated in a fixed low-rank adapter, nor do they yield an objective one can simply train with. What is missing is a regularizer

that is *derived* from a forgetting bound carrying an explicit remainder, that weights protected directions by their curvature rather than uniformly, and that keeps the parameter budget constant across tasks. The derivation is used here as a statement of *which* quantity to penalize rather than as a numerical prediction of *how much* forgetting a setting removes: Sec. 4 audits the bound’s slack empirically instead of assuming it away, and the method’s empirical justification is the ablations in Sec. 5.3. Prior work in this journal addresses adjacent aspects — gradient-based task organization [22], gradient-norm regularization for dissimilar tasks [50], and task-adaptive routing [39] — but none of it derives the protected directions from a forgetting bound, which is the specific step taken here. FOLoRA is constructed to fill that gap, and because it weights directions by Fisher eigenvalues it contains uniform (subspace-orthogonal) protection as the special case in which all eigenvalues are equal (Sec. 4).

3. Method

3.1. Preliminaries

Low-rank adaptation (LoRA). LoRA [14] freezes a pre-trained weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and injects a trainable low-rank residual,

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA, \quad (1)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$ and $r \ll \min(d_{\text{in}}, d_{\text{out}})$. The scalar α/r absorbs the rank normalization. The forward pass of an adapted layer is

$$h = W_0 x + \frac{\alpha}{r} BAx = W_0 x + \delta, \quad \delta := \frac{\alpha}{r} BAx \in \mathbb{R}^{d_{\text{out}}}, \quad (2)$$

where δ denotes the adapter’s residual contribution.

Class-incremental continual learning (CIL). We consider a sequence of T tasks $\{\mathcal{T}_t\}_{t=1}^T$. Task $\mathcal{T}_t$ provides a training set $\mathcal{D}_t = \{(x_i, y_i)\}$ over c new classes, and the classifier head is grown to cover all classes seen so far. At test time the task identity is not available, so prediction is made over the union of all seen classes. A single shared LoRA adapter is trained sequentially; its parameter count is $r(d_{\text{in}} + d_{\text{out}})$ per layer and is *independent* of T .

3.2. Motivation: forgetting as subspace overlap

Let ξ_τ collect all *trainable* parameters at the end of task $\mathcal{T}_\tau$ — the adapter parameters θ_τ (the flattened low-rank factors of Sec. 3.3) together with the classifier-head parameters — and let $\delta\xi = \xi - \xi_\tau$ be a later task’s displacement of them. A second-order Taylor expansion of task τ ’s loss around ξ_τ gives

$$\mathcal{L}_\tau(\xi_\tau + \delta\xi) \approx \mathcal{L}_\tau(\xi_\tau) + \nabla_\xi \mathcal{L}_\tau^\top \delta\xi + \frac{1}{2} \delta\xi^\top \mathcal{F}_\tau \delta\xi, \quad (3)$$

where $\mathcal{F}_\tau = \mathbb{E}_{(x,y) \sim \mathcal{D}_\tau} [\nabla_\xi \log p(y|x) \nabla_\xi \log p(y|x)^\top]$ is the empirical Fisher information matrix of task τ over those parameters. At the end of task τ the parameters are (approximately) stationary for that task’s loss, $\nabla_\xi \mathcal{L}_\tau(\xi_\tau) \approx 0$; this is stated as an explicit assumption (Assumption 1) and its residual under the regularizer is quantified in Sec. 4. Hence the *forgetting* of task τ — the increase in its loss caused by subsequent learning — is dominated by the quadratic term $\frac{1}{2} \delta\xi^\top \mathcal{F}_\tau \delta\xi$. Forgetting is therefore driven by the alignment of the new displacement with the directions in which task τ ’s loss is sharply curved (i.e., the top eigendirections of $\mathcal{F}_\tau$). Of that quadratic form, FOLoRA acts on the part involving the adapter parameters: its regularizer penalizes alignment of the adapter displacement $\delta\theta$ with the directions in which the

adapter block of $\mathcal{F}_\tau$ is sharply curved. Sec. 4 makes this precise, including the head terms that FOLoRA does not constrain.

This observation separates FOLoRA from prior work: O-LoRA [8] treats every direction of the past low-rank subspace as equally important, whereas Eq. (3) shows that directions should be protected *in proportion to their curvature*. Protecting all directions equally over-constrains the adapter and wastes the fixed rank budget on directions that carry little information; weighting by curvature spends that budget where forgetting actually happens. We note that this is an argument about *where the budget is spent*, not a claim that weighting wins: the theory does not predict an accuracy advantage for the weighted variant (Sec. 4.6 states the caveat explicitly), and Sec. 5.3 reports what our experiments show.

3.3. FOLoRA

We now describe FOLoRA. For each adapted layer, let $\theta \in \mathbb{R}^d$ denote the flattened adapter parameters (the LoRA matrices A and B , so $d = r(d_{\text{in}} + d_{\text{out}})$), and let $g_\theta = \nabla_\theta \log p(y|x)$ denote the gradient of the log-likelihood with respect to these parameters.

Importance estimation. After finishing task $\mathcal{T}_\tau$, we estimate a per-layer *importance kernel* from the per-sample parameter-gradient autocorrelation,

$$F_\tau = \mathbb{E}_{(x,y) \sim \mathcal{D}_\tau} [g_\theta(x,y) g_\theta(x,y)^\top] \in \mathbb{R}^{d \times d}, \quad (4)$$

which is the empirical Fisher information of task τ restricted to the adapter parameters — i.e. the adapter block $\mathcal{F}_\tau^{\theta\theta}$ of the full-parameter Fisher of Sec. 4 — and which we write F_τ for short below. To keep the estimate tractable we

never materialize the full $d \times d$ matrix; we collect the per-sample gradients into a matrix (one row per sample) and retain only its top- k singular directions, i.e., a rank- k approximation of F_τ . We accumulate these kernels across tasks,

$$\bar{F} = \sum_{\tau < t} F_\tau. \quad (5)$$

Eq. (5) is the *object* the protected subspace is meant to represent. The implementation stores, per layer, a rank- k factor rather than the full F_τ , so what is actually accumulated is computed on the concatenation of the retained factors of all previous tasks; this is an incremental-PCA approximation to the sum in Eq. (5) rather than the sum itself, and it is exact only when the retained top- k subspace of the concatenated factors coincides with the union of the per-task top- k subspaces. We keep the exact form in the equations for readability and use the approximation in all experiments; the two agree in the limit $k \rightarrow d$.

Protected subspace. At the start of task $\mathcal{T}_t$, we eigendecompose the accumulated kernel $\bar{F} = \sum_j \sigma_j u_j u_j^\top$, where $\{\sigma_j\}_j$ are the eigenvalues sorted in decreasing order and $\{u_j\}$ the corresponding orthonormal eigenvectors in $\mathbb{R}^d$. We retain the top- k eigendirections as an *importance-weighted protected subspace* $\mathcal{P}_t = \{(\sigma_j, u_j)\}_{j=1}^k$.

Orthogonalization regularizer. Let θ^{t-1} denote the adapter parameter snapshot recorded after task $\mathcal{T}_{t-1}$, and let $\delta\theta = \theta - \theta^{t-1}$ be the parameter displacement accumulated while training task $\mathcal{T}_t$. During the training of task

$\mathcal{T}_t$, FOLoRA minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\theta) + \lambda \underbrace{\sum_{j=1}^k \sigma_j (u_j^\top \delta\theta)^2}_{\Omega(\delta\theta)}, \quad (6)$$

where $\lambda > 0$ controls the strength and each $u_j \in \mathbb{R}^d$ is a protected direction in the same parameter space as $\delta\theta$. The term $(u_j^\top \delta\theta)^2$ penalizes the alignment of the *displacement* $\delta\theta = \theta - \theta^{t-1}$ —rather than the absolute θ —with the protected direction u_j , weighted by the importance σ_j . Using the displacement is essential: at the start of task $\mathcal{T}_t$ we have $\delta\theta = 0$, so the regularizer vanishes and the adapter’s existing alignment with past tasks is preserved; only *new* movement into protected directions is penalized. Penalizing the absolute θ would instead push the adapter away from the past tasks’ directions and actively erase them, defeating the purpose of protection. Since the regularizer is a differentiable function of A and B , it integrates with standard backpropagation and requires no hard projection, no replay buffer, and no task identity at inference.

Why importance weighting matters. Setting all $\sigma_j = 1$ reduces Ω to an equal-weight orthogonalization, which is the geometry underlying O-LoRA-style orthogonal subspace methods (Sec. 4.6 states the relation exactly). In contrast, FOLoRA’s eigenvalue weighting (i) strongly suppresses reuse of high-curvature directions, which is where forgetting is concentrated, and (ii) leaves low-curvature directions nearly free, so the adapter can reuse them for new tasks without wasting rank. This is the mechanism by which FOLoRA spends a fixed rank budget on the directions the bound identifies as costly. We note that the theory does not by itself predict an accuracy advantage for weight-

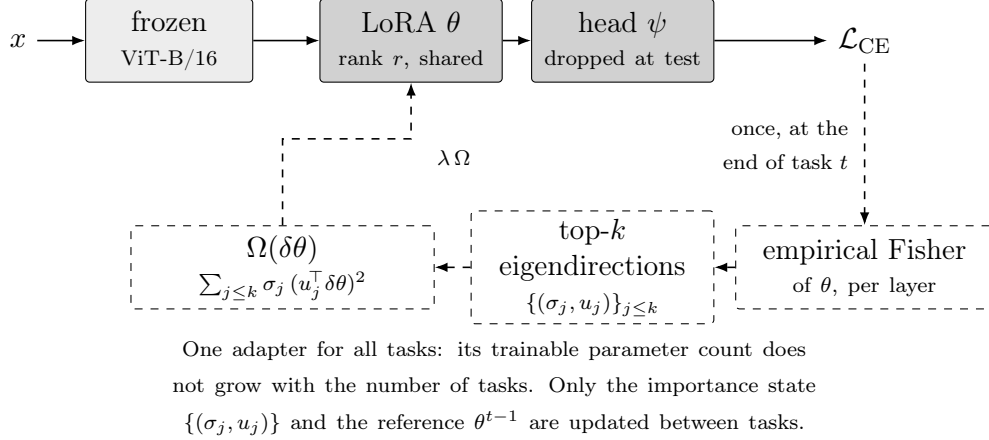


Figure 1: FOLoRA. **Solid row:** the forward pass of task t , where a single shared rank- r adapter θ is trained on a frozen ViT-B/16 backbone with the penalty $\lambda\Omega$ added to the cross-entropy loss; the head ψ is trained but discarded, since prediction is by nearest class mean in the frozen feature space. **Dashed row:** once, after task t ends, the adapter’s empirical Fisher is re-estimated per layer, its top k eigendirections are extracted, and they are accumulated into the kernel $\bar{F}$ defining Ω for the next task. The adapter’s rank, hence its trainable parameter count, is fixed for the whole sequence; only the importance state grows, so the memory cost is an importance-model cost, not a parameter cost (Table 4).

ing over equal weighting — a larger penalty family is not automatically a better one, and Sec. 4.6 says so explicitly; whether it helps in practice is an empirical question, and Sec. 5.3 reports what our experiments show.

Fig. 1 shows the procedure end to end: the forward pass of one task on the solid row, and the one-time update that produces the penalty for the next task on the dashed row. Two properties of that picture are the ones the rest of the paper relies on: the adapter and its parameter count are shared across all tasks, so nothing in the solid row grows with T ; and the quantity that is updated between tasks is the importance state $\{(\sigma_j, u_j)\}$,

Algorithm 1 FOLoRA for class-incremental continual learning

- 1: **Input:** frozen backbone; LoRA rank r ; strength λ ; budget k .
 - 2: Initialize shared LoRA adapters $\Delta W^{(l)} = \frac{\alpha}{r} B^{(l)} A^{(l)}$ for each adapted layer l .
 - 3: $\bar{F}^{(l)} \leftarrow \mathbf{0}$ for all l .
 - 4: **for** each task $\mathcal{T}_t$, $t = 1, \dots, T$ **do**
 - 5: **if** $t > 1$: eigendecompose $\bar{F}^{(l)} = \sum_j \sigma_j u_j u_j^\top$, keep top- k $\{(\sigma_j, u_j)\}$.
 - 6: Train adapter parameters $\theta^{(l)}$ and the grown classifier head by minimizing $\mathcal{L}_{\text{CE}} + \lambda \sum_l \sum_j \sigma_j (u_j^\top (\theta^{(l)} - \theta_{t-1}^{(l)}))^2$.
 - 7: Record adapter snapshot $\theta_t^{(l)} \leftarrow \theta^{(l)}$.
 - 8: Estimate $F_t^{(l)} = \mathbb{E}[g_\theta g_\theta^\top]$ on $\mathcal{D}_t$; update $\bar{F}^{(l)} \leftarrow \bar{F}^{(l)} + F_t^{(l)}$.
 - 9: **end for**
 - 10: **Output:** a single shared adapter with fixed parameter budget.
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so the memory FOLoRA spends is an importance-model cost rather than a parameter cost (quantified in Table 4). The same procedure is given in pseudocode as Algorithm 1.

3.4. Relation to prior methods

- **EWC** [3] penalizes per-parameter quadratic displacement $\sum_i F_{ii}(\theta_i - \theta_i^*)^2$ using the *diagonal* Fisher. FOLoRA instead uses the adapter block of the second-order expansion and applies it as a rank- k subspace projection rather than a pointwise penalty. The distinction is not that one is exact and the other is not — both truncate the curvature, EWC to its diagonal and FOLoRA to its top- k spectrum — but that they discard different things: EWC keeps all d coordinates and drops ev-

ery correlation among them, whereas FOLoRA keeps the k directions of largest curvature with their correlations intact and drops the rest. Which truncation suits a fixed-rank adapter is an empirical question, answered for this setting by the comparison in Sec. 5.2.

- **O-LoRA** [8] allocates a new adapter per task and initializes it orthogonally to previous tasks’ input subspaces, treating all directions equally; FOLoRA keeps a *single* fixed-budget adapter and protects directions in proportion to their Fisher importance.
- **GPM** [44] and related null-space methods project gradients onto a null space; FOLoRA realizes a *soft*, importance-weighted orthogonal projection that avoids the rank saturation of hard projections.

4. Theoretical Analysis

We derive a second-order bound on the forgetting of a previously learned task under sequential low-rank adaptation, with an explicit remainder term, and show that the FOLoRA regularizer minimizes the leading term of that bound. We then relate the bound to the equal-weight orthogonalization used by O-LoRA-style methods.

We present the bound as a *design principle rather than a numerical predictor*. We do not claim it is tight, and we never use it to predict how much forgetting a given λ removes. Sec. 4.4 audits the three quantities the expansion rests on — the cubic remainder, the stationarity residual, and the calibration of the Fisher estimate — and reports that the first two are the same order as the quadratic term rather than negligible, and that the third is

not a calibrated surrogate for the measured curvature at the operating point we probe. What the theory does establish, and what the method is built on, is structural: the leading term of the expansion is a curvature-weighted projection of the adapter displacement onto the past tasks’ eigendirections, and Eq. (15) shows that FOLoRA’s penalty is the Lagrangian form of that quantity. The *empirical* case for the method is made separately, by the $\lambda=0$ anchor in Sec. 5.3, and not by the bound.

4.1. Setup and assumptions

Let $\theta_\tau \in \mathbb{R}^d$ denote the *adapter* parameters of one adapted layer after learning task $\mathcal{T}_\tau$ — the entries of that layer’s low-rank factors A, B , flattened, so that $d = r(d_{\text{in}} + d_{\text{out}})$ — and let ψ_τ denote the classifier-head parameters at that point. Every adapted layer is treated identically, so we carry the analysis for a single layer and drop the layer index; the total adapter term is the sum over layers of the per-layer terms derived below. We write $\xi_\tau = (\theta_\tau, \psi_\tau)$ for the full trainable vector. For a later task $\mathcal{T}_t$ ($t > \tau$), let

$$\delta\theta^\tau = \theta_t - \theta_\tau, \quad \delta\psi^\tau = \psi_t - \psi_\tau, \quad \delta\xi^\tau = (\delta\theta^\tau, \delta\psi^\tau) = \xi_t - \xi_\tau \quad (7)$$

be the adapter, head and full-parameter displacements accumulated *between* τ and t , respectively. We measure the forgetting of task τ by the increase of its empirical loss,

$$\Delta\mathcal{L}_\tau = \mathcal{L}_\tau(\xi_t) - \mathcal{L}_\tau(\xi_\tau). \quad (8)$$

Our analysis is built on the following assumptions, which we state explicitly because each one is an approximation that the bound inherits.

Assumption 1 (Task stationarity). ξ_τ is a stationary point of the unregularized task loss, $\nabla_\xi \mathcal{L}_\tau(\xi_\tau) = 0$.

Assumption 2 (Hessian regularity). $\mathcal{L}_\tau$ is twice continuously differentiable on the segment $[\xi_\tau, \xi_t]$, and its Hessian is Lipschitz there with constant L_3 : $\|H_\tau(\xi) - H_\tau(\xi_\tau)\|_2 \leq L_3\|\xi - \xi_\tau\|$.

Assumption 3 (Fisher approximation). *The Hessian at the stationary point is approximated by the empirical Fisher, $H_\tau(\xi_\tau) \approx \mathcal{F}_\tau$, assumed rather than derived, following the standard practice of the EWC family [3, 4]. We state it as an assumption because the Gauss–Newton equivalence holds for the true Fisher at a maximum-likelihood point under a correctly specified model [66], whereas the empirical Fisher estimated here from a finite per-sample budget is in general a different matrix and can deviate substantially from the Hessian outside that regime [67]. The consequences of this substitution are not absorbed silently: they appear explicitly as ε_F in Eq. (9), which is by construction the exact error of the substitution at the given displacement.*

Defining ε_F as the *exact* error rather than an upper bound on it has a consequence worth stating: Eq. (9) does not actually require Assumption 3 in order to be valid — since $\Delta\mathcal{L}_\tau = \frac{1}{2}\delta\xi^{\tau\top}\mathcal{F}_\tau\delta\xi^\tau + \frac{1}{2}\delta\xi^{\tau\top}(H_\tau(\xi_\tau) - \mathcal{F}_\tau)\delta\xi^\tau + \text{remainder}$, the inequality holds for the empirical Fisher as computed, independently of its quality. The assumption governs the *usefulness* of the bound rather than its validity: it is what licenses treating ε_F as small and $\mathcal{F}_\tau$ as an object worth regularizing against.

Assumption 1 is the same assumption used by EWC [3] and GPM [44]. It applies to the *unregularized* task loss: FOLoRA’s training objective is $\mathcal{L}_{\text{CE}} + \lambda\Omega$, so at the end of task $\mathcal{T}_\tau$ we have $\nabla_\xi\mathcal{L}_{\text{CE}}(\xi_\tau) = -\lambda\nabla_\xi\Omega(\theta_\tau)$ rather than exactly zero. We discuss the size of this deviation in Remark 2 and quantify it empirically in Sec. 5.4.

4.2. A second-order forgetting bound

Theorem 1 (Second-order forgetting bound). *Under Assumptions 1–2, the forgetting of task $\mathcal{T}_\tau$ after learning task $\mathcal{T}_t$ satisfies*

$$\Delta\mathcal{L}_\tau \leq \frac{1}{2} \delta\xi^{\tau\top} \mathcal{F}_\tau \delta\xi^\tau + \frac{L_3}{6} \|\delta\xi^\tau\|^3 + \varepsilon_F, \quad (9)$$

where $\varepsilon_F = \frac{1}{2} |\delta\xi^{\tau\top} (H_\tau(\xi_\tau) - \mathcal{F}_\tau) \delta\xi^\tau|$ is the absolute error incurred by replacing the Hessian with the Fisher (Assumption 3), and $\mathcal{F}_\tau = \mathbb{E}_{(x,y) \sim \mathcal{D}_\tau} [g_\xi g_\xi^\top]$, with $g_\xi \equiv \nabla_\xi \log p_\xi(y|x)$, is the empirical Fisher of task τ over all trainable parameters.

Proof. Taylor-expanding $\mathcal{L}_\tau$ at ξ_τ along the displacement $\delta\xi = \xi_t - \xi_\tau$ with the integral form of the remainder gives

$$\mathcal{L}_\tau(\xi_t) = \mathcal{L}_\tau(\xi_\tau) + \nabla\mathcal{L}_\tau(\xi_\tau)^\top \delta\xi + \int_0^1 (1-s) \delta\xi^\top H_\tau(\xi_\tau + s\delta\xi) \delta\xi ds. \quad (10)$$

By Assumption 1 the linear term vanishes. Decomposing the integral,

$$\Delta\mathcal{L}_\tau = \frac{1}{2} \delta\xi^\top H_\tau(\xi_\tau) \delta\xi + \int_0^1 (1-s) \delta\xi^\top [H_\tau(\xi_\tau + s\delta\xi) - H_\tau(\xi_\tau)] \delta\xi ds. \quad (11)$$

The integrand is bounded by $(1-s) \|\delta\xi\|^2 \|H_\tau(\xi_\tau + s\delta\xi) - H_\tau(\xi_\tau)\|_2 \leq (1-s) s L_3 \|\delta\xi\|^3$ by Assumption 2, and $\int_0^1 (1-s)s ds = 1/6$, which bounds the remainder by $\frac{L_3}{6} \|\delta\xi\|^3$. Substituting the Fisher approximation (Assumption 3) into the leading term yields Eq. (9). $\square$

Isolating the adapter block. Because $\xi = (\theta, \psi)$ is a concatenation, the leading term of Eq. (9) splits into an adapter block, a head block and a cross term. Writing $\mathcal{F}_\tau$ in the same block structure,

$$\frac{1}{2} \delta\xi^{\tau\top} \mathcal{F}_\tau \delta\xi^\tau = \underbrace{\frac{1}{2} \delta\theta^{\tau\top} F_\tau \delta\theta^\tau}_{\text{adapter block}} + \underbrace{\delta\theta^{\tau\top} \mathcal{F}_\tau^{\theta\psi} \delta\psi^\tau + \frac{1}{2} \delta\psi^{\tau\top} \mathcal{F}_\tau^{\psi\psi} \delta\psi^\tau}_{\text{head / cross terms}}, \quad F_\tau := \mathcal{F}_\tau^{\theta\theta}, \quad (12)$$

so that Eq. (9) becomes

$$\Delta\mathcal{L}_\tau \leq \underbrace{\frac{1}{2}\delta\theta^{\tau\top}F_\tau\delta\theta^\tau}_{\text{adapter term}} + \underbrace{\frac{L_3}{6}\|\delta\xi^\tau\|^3}_{\text{higher order}} + \varepsilon_F + R_{\text{head}}, \quad (13)$$

where R_{head} collects the two head/cross terms of Eq. (12) and F_τ is the adapter block of $\mathcal{F}_\tau$ — the empirical Fisher restricted to the adapter parameters of one layer, of size $d \times d$. FOLoRA constrains the first term of Eq. (13) and nothing else; we carry R_{head} explicitly rather than dropping it, because the classifier head is trained between τ and t and its contribution to $\Delta\mathcal{L}_\tau$ is not zero.

Three approximations are made in *estimating* F_τ , as opposed to in the bound itself, and we state all three. First, the estimate is a rank- k truncation, as described in Sec. 3. Second, it is computed per adapted layer and treated as block-diagonal across layers: correlations between the parameters of different adapted layers are not modelled, so the $d \times d$ block above is one diagonal block of the full adapter Fisher of dimension $Lr(d_{\text{in}} + d_{\text{out}})$ for L adapted layers. Third, the weights are on the scale of an *unnormalized* estimate. The per-direction weight σ_j^2 carried by the released implementation is an eigenvalue of $G^\top G$ for the matrix G of per-sample adapter gradients accumulated over the Fisher budget, and therefore grows with the number of samples in that budget (300 at our default), whereas the diagonal Fisher of the EWC baseline we compare against is explicitly averaged over its samples. A numerical value of λ is consequently meaningful only relative to the convention its own method uses: the λ grids of Sec. 5 are read in that sense, and we do not compare λ across the two families on an absolute scale. All three are properties of the estimator; the bound is stated for the exact F_τ , so

none weakens Theorem 1, but all three are part of what a reader comparing the paper to the released code should expect to find. The decomposition is useful only if the adapter term dominates, which holds when the displacement is small. In our setting this is not merely a hope: the adapter has a fixed rank $r = 16$ per layer and is trained under an explicit penalty on its displacement, so $\|\delta\theta^\tau\|$ is bounded by the geometry of the problem rather than growing with the number of tasks. Whether the higher-order term is in fact negligible at that displacement is a question about the data, not about the algebra, and we measure it rather than assume it: Sec. 4.4 finds it to be the *same order* as the quadratic term, which is why Eq. (13) is described throughout as an upper bound with quantified slack rather than as a tight characterization of forgetting.

Eq. (13) is a bound on the *loss* of task τ , and in that bound the classifier head contributes through R_{head} . Our evaluation, however, does not use the head: under the NCM protocol (Sec. 5.1) the head is discarded at test time and prediction is made by nearest class mean in the frozen feature space, so the adapter is the *only* pathway by which a task’s test accuracy can be affected after it has been learned. The bound’s scope and the evaluation protocol therefore coincide in the sense that matters for the reported metric, even though they differ in the loss that the bound is stated for.

Remark 1 (Scope: one reference point). *Summing Eq. (13) over all previous tasks $\tau < t$ requires a displacement $\delta\theta^\tau$ per task τ , i.e. a snapshot of the adapter parameters at every past task. FOLoRA stores a single reference θ^{t-1} , following the same single-reference approximation as EWC [3], and replaces $\delta\theta^\tau$ by $\delta\theta = \theta_t - \theta^{t-1}$. The induced error is $\theta^{t-1} - \theta^\tau$, i.e. exactly*

the displacement accumulated between tasks τ and $t - 1$ — which is itself penalized by the same regularizer at every intermediate task. Writing R for the head terms R_{head} summed over τ , the bound we actually minimize is

$$\begin{aligned} \sum_{\tau < t} \Delta \mathcal{L}_\tau &\leq \frac{1}{2} \delta \theta^\top \left(\sum_{\tau < t} F_\tau \right) \delta \theta + (t - 1) \left(\frac{L_3}{6} \|\delta \xi^{\max}\|^3 + \varepsilon_F \right) \\ &+ R + \mathcal{O}(\|\theta^{t-1} - \theta^\tau\|^2), \end{aligned} \quad (14)$$

where we identify $\sum_{\tau < t} F_\tau$ with the accumulated importance kernel $\bar{F}$ of Eq. (5). The scope of the theorem is thus “forgetting relative to the most recent snapshot”, and the statement degrades gracefully rather than being exact for all historical tasks.

Remark 2 (Deviation from stationarity under the regularizer). *Because FOLoRA minimizes $\mathcal{L}_{CE} + \lambda \Omega$, the stationarity Assumption 1 holds only approximately, with residual gradient $-\lambda \nabla_\xi \Omega(\theta_\tau)$ — nonzero only in the adapter block, since Ω does not depend on the head. Writing $\Omega(\delta \theta) = \sum_j \sigma_j (u_j^\top \delta \theta)^2$ gives $\nabla_\theta \Omega(\theta_\tau) = 2 \bar{G}^\top \bar{G} \delta \theta$ where $\bar{G}$ stacks the accumulated Fisher directions. The residual therefore vanishes proportionally to the displacement itself: in directions where protection is effective ($\delta \theta \rightarrow 0$) the assumption is self-consistently satisfied, while in weakly protected directions it degrades linearly in λ . This provides a theoretical counterweight to increasing λ beyond the range where the displacement is small, and is consistent with the ablation in Sec. 5.3: at $k=16$, where the penalty acts on fewer directions, average accuracy peaks at $\lambda \in \{3, 10\}$ and then declines through $\lambda = 100$ before saturating by $\lambda = 300$. At $k=64$ the same decline is not visible over the range we probed, and we do not claim it there.*

4.3. FOLoRA minimizes the bound

Using the eigendecomposition $\bar{F} = \sum_j \sigma_j u_j u_j^\top$, the leading term of Eq. (14) becomes

$$\frac{1}{2} \delta\theta^\top \bar{F} \delta\theta = \frac{1}{2} \sum_j \sigma_j (u_j^\top \delta\theta)^2, \quad (15)$$

where $\delta\theta \in \mathbb{R}^d$ is the adapter-parameter displacement and each $u_j \in \mathbb{R}^d$ is a unit eigenvector of $\bar{F}$, so every inner product above is between vectors of the same dimension. Equation (15) shows that the dominant term of the forgetting bound is a sum of squared projections of the displacement onto the eigendirections of the accumulated Fisher, weighted by their importance σ_j . Here σ_j denotes an eigenvalue of $\bar{F}$; the implementation stores each direction scaled by the corresponding singular value $s_j = \sqrt{\sigma_j}$ of the per-sample gradient matrix, so that the squared norm it penalizes carries weight σ_j exactly as written. The FOLoRA regularizer in Eq. (6),

$$\Omega(\delta\theta) = \sum_j \sigma_j (u_j^\top \delta\theta)^2, \quad \delta\theta = \theta - \theta^{t-1}, \quad (16)$$

is exactly the penalty term of the Lagrangian relaxation of the constraint that the leading term of the bound remain below a tolerance $\epsilon > 0$,

$$\delta\theta^\top \bar{F} \delta\theta \leq \epsilon, \quad (17)$$

since relaxing that constraint with multiplier λ returns $\mathcal{L}_{\text{CE}} + \lambda\Omega$. Minimizing Ω therefore drives $|u_j^\top \delta\theta|$ toward zero for the largest σ_j , which in turn minimizes the dominant terms of Eq. (15). We therefore interpret FOLoRA as the algorithm that *directly minimizes the leading term of a second-order bound on catastrophic forgetting*, with λ controlling the stability–plasticity trade-off and k controlling the rank of the protected subspace.

4.4. How tight is the bound?

Theorem 1 is an upper bound, and everything above uses only its *structure*. This subsection reports what the bound’s own terms evaluate to on a trained run, because two of the approximations it inherits turn out to be violated by amounts that change how the bound should be read.

All numbers below come from an in-training probe that, at each task boundary of a 20-task run, fits the expansion of the accumulated past-task loss along the *realized* adapter displacement, $\Delta\mathcal{L}(\alpha\delta\theta) = a\alpha + b\alpha^2 + c\alpha^3 + \mathcal{O}(\alpha^4)$, over $\alpha \in [0, 1.5]$. Measuring the nonlinearity along the realized displacement, rather than bounding L_3 , is what makes the remainder checkable at all: L_3 is not recoverable from any stored checkpoint, whereas a , b and c are. We report a and c as ratios to $|b|$, so that “is this correction small?” is read directly off the number. The probe run is FOLoRA at $\lambda=10$, $k=16$, seed 0 on Split CIFAR-100, giving 19 usable boundaries. It is a single configuration, and we do not have the seed spread needed to say how much of what follows is run-to-run variation; the numbers should be read as one operating point, not as a distribution over the method.

(i) *The remainder is not negligible.* The ratio $|c|/|b|$ has median 0.163, minimum 0.010 and maximum 2.46, with 5 of the 19 boundaries above 0.30. Under the decision rule frozen before these numbers were computed (median ≤ 0.10 : keep the negligibility claim; $0.10 < \text{median} \leq 0.30$: weaken it; > 0.30 : drop it), this lands in the middle band, and the honest reading is the one the frozen rule prescribes: the cubic term is *of the same order as, but does not dominate*, the quadratic term. The tail is the part that matters — at the worst boundary the quantity the bound calls a remainder is $2.5\times$

the quantity FOLoRA minimizes.

(ii) *The stationarity residual is not small either.* The ratio $|a|/|b|$ has median 0.99 and maximum 23.1. Assumption 1 is stated for the *unregularized* loss, so at a regularized endpoint the residual is $-\lambda \nabla \Omega$, and Remark 2 argued that it should vanish proportionally to the displacement. It does not vanish fast enough to be dropped: at the realized displacement the linear term is comparable to the quadratic one, so the “quadratic plus corrections” reading of the expansion is not numerically available at this λ .

(iii) *The Fisher estimate is not a calibrated surrogate for the curvature.* The ratio between the term the bound prescribes, $\frac{1}{2} \delta \theta^\top \bar{F} \delta \theta$, and the quadratic coefficient b fitted from the same measurement has median 2.8×10^{-5} : the finite-sample, rank- k , block-diagonal estimate of F_τ underestimates the measured curvature of the past loss along the realized displacement by four to five orders of magnitude. It is worth being precise about what this does and does not say. It does *not* contradict Theorem 1, which is stated for the exact F_τ ; it says that Assumption 3 is far from innocuous at this operating point, which is precisely the regime the empirical-Fisher literature warns about [67]. It is also, in part, the intended behaviour seen from the other side: the displacement is close to orthogonal to the top- k Fisher subspace *by construction*, since suppressing that alignment is what the penalty does, so the Fisher-weighted quadratic form is small along it. Both readings agree on the operative point: the bound’s leading term is not a quantitative predictor of forgetting in this setting.

What this licenses. The bound is used here as a *design principle*. It identifies which quantity a regularizer should suppress — the curvature-weighted projection of the adapter displacement onto the past tasks’ protected directions — and Eq. (15) shows that FOLoRA’s penalty is exactly the Lagrangian form of that quantity. We make no claim that minimizing it produces a predictable reduction in forgetting, and Sec. 5 does not test such a claim. The empirical case for FOLoRA is made by ablation instead, and it is a clean one: removing Ω from the objective and changing nothing else takes Split CIFAR-100 accuracy from 78.32 to 69.59 ($n=5$, paired $p = 0.0005$, negative in all five seeds), i.e. down to the level of unprotected sequential fine-tuning (69.31, $p = 0.53$). The entire effect of the method is carried by the term the bound identifies, which is the sense in which the derivation is load-bearing — as motivation for *what* to penalize, not as a guarantee of *how much* it helps.

4.5. Relation to prior forgetting bounds

Second-order analyses of catastrophic forgetting are well established for models trained from scratch. Nguyen et al. [51] show that forgetting is driven by the alignment between successive tasks’ loss landscapes and derive a worst-case bound in terms of the curvature of the old task; Evron et al. [52] give tight bounds for linear networks showing that forgetting is governed by the overlap between the old task’s input distribution and the directions in which the new task changes the weights; Karakida and Akaho [53] analyse learning curves for continual learning using NTK overlap matrices; and Doan et al. [54] quantify forgetting through the NTK overlap between tasks. Closer in spirit, orthogonal-gradient descent [46, 43] admits generalization bounds obtained by constraining updates to the null space of previous task gradients.

Our analysis differs from this line in where it is instantiated. Prior results bound forgetting for *full* parameter updates of models trained from scratch, and are used mainly to explain forgetting. We instantiate the bound in the *low-rank adapter subspace* of a frozen pre-trained backbone, where the displacement is supported on a $r(d_{\text{in}} + d_{\text{out}})$ -dimensional subspace per layer, and we use the bound constructively: its dominant term is exactly the quantity a subspace-regularizer can minimize, which yields the FOLoRA objective. The resulting penalty differs from null-space projection [44, 46] in being importance-weighted and soft rather than hard, so that the fixed rank budget is spent on the directions with the largest curvature.

4.6. Connection to equal-weight orthogonalization

If all eigenvalues are set to $\sigma_j = 1$, the leading term of the bound, Eq. (15), becomes $\frac{1}{2}\|\bar{U}^\top \delta\theta\|^2$, where $\bar{U} = [u_1, \dots, u_k]$ is the orthonormal basis of the retained protected subspace — the equal-weight geometry underlying O-LoRA-style orthogonal subspace methods. The eigenvalue-weighted family therefore contains the equal-weight geometry as the special case $\sigma_j \equiv 1$, and the two coincide exactly when the spectrum is flat over the protected subspace; on our checkpoints it is not flat, so the weighted family is the strictly larger one as a family of penalties.

A larger family is not by itself an accuracy claim, and one tempting overclaim is worth pre-empting here: because every coefficient in Eq. (15) is non-negative, weighting by σ_j does *not* reduce the value of the bound at a fixed displacement — the weighted coefficients can exceed 1. The claim is the weaker and well-defined one that the weighted penalty attains a *lower value of the bound after both objectives are optimized* for a given protected rank k ,

by spending the fixed rank budget where the curvature is largest. Whether that is realized in practice is an empirical question; Sec. 5.3 characterizes the weighted family over λ and k , and the equal-weight special case $\sigma_j \equiv 1$ is its natural control.

5. Experiments

5.1. Setup

We use a ViT-B/16 backbone pre-trained on ImageNet-1k, frozen throughout, with LoRA adapters (rank $r = 16$, $\alpha = 16$) attached to the MLP down-projection of every transformer block (12 adapters in total). The classifier head is a single linear layer grown as new classes arrive. All methods share the identical backbone and head; they differ only in the adapter regularization.

Datasets. Split CIFAR-100 (20 tasks $\times$ 5 classes) and Split ImageNet-R (20 tasks $\times$ 10 classes), both under the class-incremental protocol (task identity unavailable at test time).

Baselines. (i) *Seq-LoRA*: sequential fine-tuning of the shared adapter with no protection (lower bound); (ii) *EWC-LoRA*: diagonal Fisher on the LoRA parameters [3]; (iii) *O-LoRA*: per-task adapters with orthogonal initialization [8]; selected per input by key–query matching [6]; (v) *CODA-Prompt*: a set of prompt components combined by attention [7]; (vi) *InfLoRA*: a per-task subspace projected to interfere minimally with the previous ones [19]; (vii) *FOLoRA* (ours).

Training. AdamW with learning rate 10^{-3} , batch size 32, and bf16 mixed precision. We train the adapter-based methods (Seq-LoRA, EWC-LoRA, O-

LoRA, FOLoRA) for 5 epochs per task, which avoids over-fitting and keeps the empirical Fisher meaningful (Sec. 5.4); the prompt-based methods (L2P, CODA-Prompt) are trained for 20 epochs per task, their optimal setting since their key-based prompt selection benefits from more epochs. Prompt geometry is a substantive choice rather than a detail — CODA-Prompt’s contribution is a *large* pool of components combined by attention — so we use each prompt method’s own geometry (L2P: pool 20, length 5; CODA-Prompt: pool 100, length 8) and report CODA-Prompt at its faithful geometry. We also ran CODA-Prompt with L2P’s geometry (pool 20, length 5) at 20 epochs; it scores 75.83 on CIFAR-100 against the faithful 74.71. We report the faithful configuration, i.e. the *lower* of the two numbers we measured, so the CODA-Prompt margin quoted below is the smaller of the two available to us. For FOLoRA we fix the main configuration by a grid search and then do not retune it: $\lambda \in \{1, 3, 10\}$ at $k=64$ and $k \in \{16, 64, 128\}$ at $\lambda=3$, plus $\lambda=0$ (penalty removed, everything else unchanged) as the ablation control. The main configuration $k=64$, $\lambda=3$ was fixed *before* the extra seeds of the top-up landed and is reported as such throughout; $\lambda=10$ scores marginally higher on the five seeds it was run on (78.24 against 77.87) and appears as a grid point in Table 3, not as the main configuration. The importance kernel is estimated from up to 300 per-sample gradients per task. EWC’s regularization strength is likewise tuned on CIFAR-100: $\lambda \in \{1, 3, 10, 30\}$ at five seeds, and a wider grid $\lambda \in \{100, 300, 1000, 3000\}$ at ten seeds that was run on both benchmarks ($\lambda=3000$ was not run on ImageNet-R). In Table 1 we report each benchmark’s *best-scoring* EWC point from that grid — $\lambda=300$ on CIFAR-100 and $\lambda=1000$ on ImageNet-R — i.e. we deliberately take the

strongest configuration of the baseline *on each benchmark*, and we report the whole curve, including the pre-registered baseline $\lambda=100$, in Table 2.

Metrics and evaluation protocol. We follow the frozen-feature nearest-class-mean (NCM) protocol standard in this literature [6, 7, 19]: after the final task we L2-normalize the CLS feature of the adapted backbone and classify by the nearest class mean computed from the training features of each task seen so far, with each task’s means computed from that task’s own training split. This is the protocol under which L2P, CODA-Prompt and InfLoRA are reported, which makes the comparison directly interpretable; it also removes the classifier head from the measurement, so the reported numbers isolate the quality of the adapted representation rather than the growth of the head. We report average accuracy over all tasks after the final task ($\overline{\text{ACC}}$), the CIL–TIL gap (GAP), and average incremental accuracy. GAP is the mean drop, in points, when a task’s test set is classified against all seen classes rather than against that task’s own classes, under the same final-model feature extraction: it measures how far a method’s decision rule is from relying on task identity, and lower is better. Because every method is run on the same seed set—and therefore sees the same class order and initialization—we assess the significance of differences between methods with a paired t -test over the seeds common to both. Seed budgets are *not* equal across methods and we do not pool them: EWC-LoRA and FOLoRA have $n = 10$ on both benchmarks; Seq-LoRA and O-LoRA $n = 10$ on CIFAR-100 and $n = 5$ on ImageNet-R; InfLoRA 5/3; L2P 5/5; CODA-Prompt 5/3. SimpleCIL is a deterministic floor ($n = 1$, no variance) and is not tested. Every p -value below states the common-seed count it was computed on,

and Table 1 repeats the budget for each method. Except for the single pre-registered verdict discussed in Sec. 5.2, these tests are reported *without* correction for multiple comparisons. Each family—the baseline contrasts of Table 1, the EWC λ grid of Table 2—is tested at the nominal level, so family-level statements should be read with that in mind. We report the individual p -values rather than corrected ones so that a reader who prefers a correction can apply one directly; the two claims the argument leans on most heavily are not marginal ($p = 0.0005$ for the $\lambda=0$ ablation of Sec. 5.3 and $p = 0.0002$ for the pre-registered ImageNet-R comparison).

5.2. Main results

The protocol-independent facts hold across both tables: all methods share one frozen backbone and one classifier head, and the trainable parameter count of FOLoRA, Seq-LoRA, EWC-LoRA, L2P and CODA-Prompt is fixed across tasks, whereas O-LoRA and InfLoRA allocate a new adapter per task and therefore grow linearly with the number of tasks (Table 4). Fig. 2 and Fig. 3 show the per-task trajectories that the final-task averages of Table 1 collapse. On CIFAR-100 all six methods start within five points of each other (94.2 to 98.6) while their final accuracies span 64.2 to 78.3; FOLoRA and EWC-LoRA in particular are indistinguishable throughout, their paired difference ranging over only 0.98 points across the whole sequence and never exceeding 0.92 in either direction. On ImageNet-R the ordering is already visible at the first point (FOLoRA 92.0, O-LoRA 72.0), and FOLoRA’s lead over EWC-LoRA stays small throughout rather than widening: +1.18 at the first task, -0.37 at the fifth, $+0.90$ at the twentieth, a spread of 1.54 points — which is why we do not read that accuracy difference as an advantage

Table 1: Class-incremental results under the frozen-feature nearest-class-mean protocol (mean $\pm$ std over seeds; $\overline{\text{ACC}}$ higher is better). GAP is the CIL–TIL gap: the mean drop, in points, when a task’s test set is classified against all seen classes rather than against that task’s own classes, under a single final-model feature extraction. Lower is better — it means the decision rule relies less on task identity. EWC-LoRA is shown at the best of its tuned points on each benchmark — $\lambda=300$ (CIFAR-100); $\lambda=1000$ (ImageNet-R) — i.e. the strongest configuration of the baseline, with its full λ curve, including the pre-registered judging baseline $\lambda=100$, in Table 2. The accuracy comparison is parity at every point tested *except* ImageNet-R at $\lambda=100$ ($+2.85$, $p = 0.0002$), where the paired difference reaches significance. On CIFAR-100 the difference changes sign across the grid ($+0.65$ at $\lambda=100$ to -0.14 at $\lambda=300$), so the ordering of the two methods is not stable across the baseline’s own tuned points. The CIL–TIL gap comparison is mixed: EWC-LoRA has the *smaller* CIL–TIL gap at CIFAR-100 $\lambda=1000$, CIFAR-100 $\lambda=3000$; FOLoRA has the smaller gap at ImageNet-R $\lambda=100$, ImageNet-R $\lambda=300$. Both directions are detailed in Table 2; the points not named there are not significant. Seed budgets are *not* equal and are stated per method: EWC-LoRA and FOLoRA $n=10$ on both benchmarks; Seq-LoRA and O-LoRA 10 (CIFAR-100) / 5 (ImageNet-R); InfLoRA 5/3; L2P 5/5; CODA-Prompt 5/3; SimpleCIL is the deterministic frozen-feature floor ($n=1$, no variance, not tested). Bold marks the best value *among the rows shown here*; where the best is tied no value is bolded.

Method	Split CIFAR-100		Split ImageNet-R	
	$\overline{\text{ACC}}$	GAP	$\overline{\text{ACC}}$	GAP
Seq-LoRA	69.31 $\pm$ 1.92	9.92 $\pm$ 0.58	56.58 $\pm$ 1.49	8.83 $\pm$ 0.70
EWC-LoRA	78.00$\pm$0.54	7.35$\pm$0.37	66.33 $\pm$ 0.61	7.00$\pm$0.30
O-LoRA	64.79 $\pm$ 2.08	9.73 $\pm$ 0.55	44.79 $\pm$ 1.58	8.25 $\pm$ 0.40
L2P	73.65 $\pm$ 0.81	8.12 $\pm$ 0.37	59.47 $\pm$ 0.43	8.02 $\pm$ 0.36
CODA-Prompt	74.71 $\pm$ 0.71	7.71 $\pm$ 0.55	59.31 $\pm$ 0.22	7.83 $\pm$ 0.42
InfLoRA	70.14 $\pm$ 0.26	8.87 $\pm$ 0.39	50.00 $\pm$ 0.48	7.53 $\pm$ 0.44
SimpleCIL (floor)	70.31 $\pm$ 0.00	8.48 $\pm$ 0.00	49.71 $\pm$ 0.00	7.96 $\pm$ 0.00
FOLoRA (ours)	77.87 $\pm$ 0.90	8.42 $\pm$ 0.59	66.37$\pm$0.77	7.27 $\pm$ 0.49

Table 2: EWC-LoRA regularization sensitivity. Each λ is compared against FOLoRA with a paired t -test over the seeds common to both; n is that common count, *not* either side’s own budget, and it is why the FOLoRA column repeats the same value on every row of a benchmark. $\lambda=100$ is the baseline fixed in advance, before the seed top-up that closed the grid to $n=10$; $\lambda=3000$ was not run on ImageNet-R. * marks $p < 0.05$. The accuracy comparison is parity at every point tested *except* ImageNet-R at $\lambda=100$ (+2.85, $p = 0.0002$), where the paired difference reaches significance. On CIFAR-100 the difference changes sign across the grid (+0.65 at $\lambda=100$ to -0.14 at $\lambda=300$), so the ordering of the two methods is not stable across the baseline’s own tuned points. The CIL–TIL gap comparison is mixed: EWC-LoRA has the *smaller* CIL–TIL gap at CIFAR-100 $\lambda=1000$ (7.10 against 7.42, $p = 0.0400$), CIFAR-100 $\lambda=3000$ (7.03 against 7.42, $p = 0.0401$); FOLoRA has the smaller gap at ImageNet-R $\lambda=100$ (8.44 against 7.27, $p = 0.0004$), ImageNet-R $\lambda=300$ (7.67 against 7.27, $p = 0.0317$) — neither direction dominates.

Benchmark	EWC λ	n	$\overline{\text{ACC}}$			
			EWC	FOLoRA	Δ	p
CIFAR-100	100	10	77.22 $\pm$ 1.08	77.87 $\pm$ 0.90	+0.65	0.0734
CIFAR-100	300	10	78.00 $\pm$ 0.54	77.87 $\pm$ 0.90	-0.14	0.6440
CIFAR-100	1000	10	77.65 $\pm$ 0.42	77.87 $\pm$ 0.90	+0.21	0.4961
CIFAR-100	3000	10	77.40 $\pm$ 0.62	77.87 $\pm$ 0.90	+0.47	0.2507
ImageNet-R	100	10	63.52 $\pm$ 1.28	66.37 $\pm$ 0.77	+2.85	0.0002*
ImageNet-R	300	10	66.32 $\pm$ 0.62	66.37 $\pm$ 0.77	+0.05	0.9010
ImageNet-R	1000	10	66.33 $\pm$ 0.61	66.37 $\pm$ 0.77	+0.04	0.9235
ImageNet-R	3000	—	—	66.37 $\pm$ 0.77	—	—

that compounds with sequence length.

FOLoRA matches the strongest regularized baseline; it does not beat it. Under the comparison fixed in advance — a paired t -test against EWC-LoRA at $\lambda = 100$ — FOLoRA is ahead by +0.65 points on CIFAR-100 ($p = 0.0734$, which the pre-registered rule calls “matches or slightly exceeds” and *not* “outperforms”) and by +2.85 points on ImageNet-R ($p = 0.0002$, all ten per-seed differences positive), which the same rule calls a significant lead. We report that verdict, and also why it must not be read as a general advantage: λ is a hyperparameter of the *baseline*, and the comparison moves with it. On CIFAR-100 EWC-LoRA’s four tuned points span 77.22 to 78.00 while FOLoRA scores 77.87 — inside that span — with paired differences of -0.14 ($p = 0.644$) at $\lambda = 300$, $+0.21$ ($p = 0.496$) at $\lambda = 1000$ and $+0.47$ ($p = 0.251$) at $\lambda = 3000$. That is parity at every CIFAR-100 point, and EWC-LoRA’s best point sits 0.13 points *above* FOLoRA. On ImageNet-R the same structure appears with a wider spread, and the lead is confined to one end of it: $\lambda = 100$ is that benchmark’s weakest EWC point at 63.52, and it is there — and only there — that the comparison reaches significance, while at $\lambda = 300$ and $\lambda = 1000$, both roughly 2.8 points higher, the difference collapses to $+0.05$ ($p = 0.901$) and $+0.04$ ($p = 0.924$). The sign of our difference against EWC-LoRA is therefore set by which operating point of the baseline is read, so we report the whole curve (Table 2) rather than the point that reads best for us. EWC-LoRA was not run at $\lambda = 3000$ on ImageNet-R.

On the CIL–TIL gap the comparison is mixed, and we report both directions. The direction is set by the baseline’s regularization

strength, not by the method. On CIFAR-100 EWC-LoRA has the significantly *smaller* gap at the two larger strengths: the paired difference is +0.32 points at $\lambda = 1000$ ($p = 0.0400$) and +0.39 at $\lambda = 3000$ ($p = 0.0401$), with +0.07 ($p = 0.568$) at $\lambda = 300$ and -0.30 ($p = 0.071$) at the pre-registered $\lambda = 100$, where the sign runs the other way without reaching significance. On ImageNet-R the direction reverses: FOLoRA has the significantly *smaller* gap at $\lambda = 100$ (-1.17 points, $p = 0.0004$) and at $\lambda = 300$ (-0.41 , $p = 0.0317$), while at $\lambda = 1000$ the two are indistinguishable ($+0.27$, $p = 0.203$). Because average accuracy and the gap are measured under the same final feature space—a representation that separates the classes better both classifies better and loses less when the candidate set grows—the two co-vary across these methods. We therefore do not treat the gap as an independent axis of merit in either direction; we report it rather than leaving a reader to assume the favourable direction.

FOLoRA is ahead of the parameter-efficient family, and there the difference is uniform. Against L2P, CODA-Prompt, InfLoRA and O-LoRA the paired differences in average accuracy are significant on both benchmarks with every per-seed difference of the same sign: across the four baselines the mean paired difference runs from +3.61 to +13.07 points on CIFAR-100 ($p \leq 0.002$) and from +7.45 to +22.13 on ImageNet-R ($p \leq 0.003$); the upper end, +13.07, is measured against the initialization-only O-LoRA row and narrows to +9.93 against the training-time orthogonalized variant that the original paper specifies: $p = 0.012$ at one constraint strength and $p = 0.005$ at the other, both of which we ran and report (Sec. 5.4). This is the paper’s positive empirical result. It is also the comparison in which the

baselines’ seed budgets are smallest ($n = 5$, or $n = 3$ for CODA-Prompt and InfLoRA on ImageNet-R), which is why Sec. 5.1 lists each budget separately instead of implying a shared one — and why the claim is stated against the parameter-efficient family rather than against full-parameter regularization. Within that family, both O-LoRA and InfLoRA fall below the frozen-feature floor on CIFAR-100 (64.79 and 70.14 against 70.31); on ImageNet-R O-LoRA falls below it (44.79 against 49.71), while InfLoRA sits marginally above it (50.00, i.e. +0.29). Sec. 5.4 treats the O-LoRA and the CIFAR-100 InfLoRA results as a finding about task-ID-free merging at $T=20$ rather than as evidence about those methods.

5.3. Ablation study

We ablate FOLoRA’s two hyperparameters — the regularization strength λ and the number of protected directions k — on Split CIFAR-100. Table 3 reports average accuracy and the CIL–TIL gap for each setting together with a paired t -test against the pre-registered main configuration ($k=64$, $\lambda=3$).

The penalty carries the entire effect. Setting $\lambda = 0$ removes the Fisher-weighted orthogonality penalty and changes nothing else; on the five seeds common to both configurations average accuracy falls from 78.32 to 69.59 ($\Delta = +8.72$, $p = 0.0005$), i.e. to the level of unprotected sequential fine-tuning (69.31, Table 1). (The $\lambda=3$ full ten-seed mean is the 77.87 of Table 1; the two figures differ only in seed budget, not in configuration.) The benefit of FOLoRA is therefore attributable to the regularization term rather than to the shared adapter, the rank constraint or the training recipe — which is what the derivation predicts: with no penalty the protected subspace is empty and the method degenerates to Seq-LoRA.

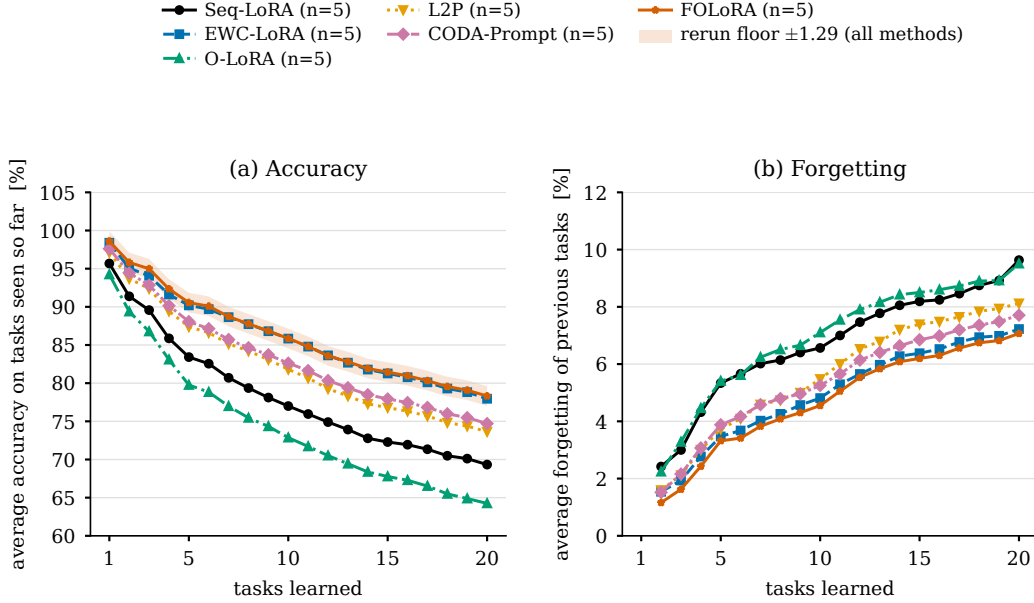


Figure 2: Average accuracy over the tasks seen so far (left) and the mean CIL–TIL gap over those tasks (right) on Split CIFAR-100, as a function of the number of tasks learned. Each curve is the mean over that method’s own available seeds (on this benchmark every method has $n=5$; the configuration per method is the one of Table 1). No smoothing or interpolation is applied — the markers are the per-task measurements. The shaded band is the run-to-run reproducibility floor of this measurement (1.29 points, Sec. 5.4); it is a property of the protocol rather than of any one method, and it applies to every curve shown. The vertical axis is bounded below at 60, within five points of the lowest value plotted, rather than at zero, so that the differences are legible; the full range is 60–105. Because the figure averages over at most five seeds per method (each method’s own) while Table 1 reports each method at its full budget, the last points differ slightly — FOLoRA ends at 78.32 here against 77.87 there — and on those five seeds FOLoRA also sits ahead of EWC-LoRA (78.32 against 77.96), the reverse of the ordering in Table 1. The two are level over the full budgets, the 0.13-point gap there lying inside the 1.29-point reproducibility floor.

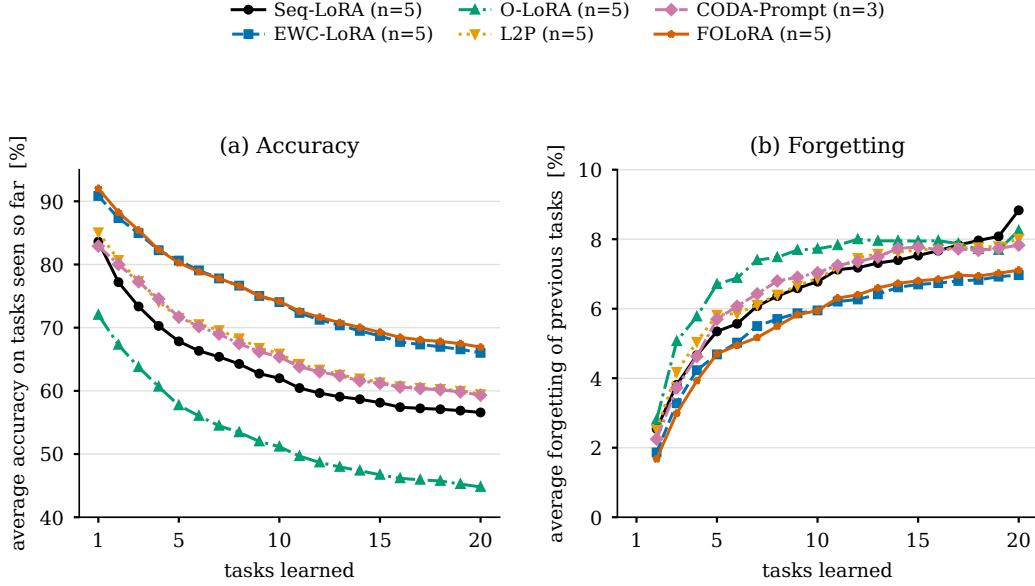


Figure 3: Average accuracy over tasks seen so far (left) and the mean CIL–TIL gap over those tasks (right) on Split ImageNet-R. Seed budgets are *not* uniform on this benchmark: CODA-Prompt was run on three seeds and every other method on five, so its curve averages three runs and the others five — we do not imply equal n . No reproducibility-floor band is drawn, because that floor was measured on CIFAR-100 only and we do not transfer it to another dataset. The vertical axis is bounded below at 40, i.e. 4.79 points under the lowest value plotted. Each method is drawn at its own seed budget — there is no subset common to all six on this benchmark, since CODA-Prompt has only three runs — so the two methods whose budget exceeds five end lower here than in Table 1: FOLoRA at 66.92 against 66.37 at ten seeds, and EWC-LoRA at 66.02 against 66.33; both keep the configuration of their row there ($k=64$ and $\lambda=1000$, respectively). The 0.31-point EWC-LoRA difference is disclosed for the same reason as FOLoRA’s: the curve is a five-seed subset, the table a ten-seed budget.

Table 3: Ablation on Split CIFAR-100. Δ and p compare each setting with the pre-registered main configuration ($k=64$, $\lambda=3$) using a paired t -test over common seeds; n is each row’s own seed budget, so Δ is computed on the intersection and need not equal the difference of the two means shown. $\lambda=0$ removes the Fisher-weighted orthogonality penalty and changes nothing else. GAP is the CIL–TIL gap of Table 1, defined there; lower is better.

Setting	n	$\overline{\text{ACC}}$	GAP	Δ vs main	p
λ at $k=64$					
$\lambda=0$	5	69.59 ± 1.55	9.53 ± 0.65	+8.72	0.0005*
$\lambda=1$	5	77.33 ± 1.13	7.56 ± 0.80	+0.99	0.1106
$\lambda=3$	10	77.87 ± 0.90	7.42 ± 0.59	(<i>main</i>)	—
$\lambda=10$	5	78.24 ± 0.64	6.87 ± 0.30	+0.08	0.8868
k at $\lambda=3$					
$k=16$	5	78.14 ± 0.68	7.32 ± 0.61	+0.17	0.7487
$k=64$	10	77.87 ± 0.90	7.42 ± 0.59	(<i>main</i>)	—
$k=128$	5	77.66 ± 0.63	7.49 ± 0.42	+0.65	0.2768

At $k=64$ the λ response is a step, not a peak, over the range we probed. Accuracy rises from 69.59 at $\lambda = 0$ to 77.33 at $\lambda = 1$ and 77.87 at $\lambda = 3$, then stays flat (78.24 at $\lambda = 10$). Only $\lambda = 0$ separates from the main configuration ($p = 0.0005$); $\lambda \in \{1, 3, 10\}$ are mutually indistinguishable (largest $p = 0.887$, smallest 0.111). We therefore do not claim an interior optimum at $k=64$ — the grid does not resolve one, and any ordering among $\lambda \in \{1, 3, 10\}$ is inside the run-to-run band of Sec. 5.4.

At $k=16$ the decline at large λ is visible. Extending the sweep on the smaller protected subspace, where the penalty acts on fewer directions, average accuracy peaks at $\lambda \in \{3, 10\}$ (78.14 and 77.90, both $n=5$) and then declines before plateauing: 75.35 at $\lambda = 100$ ($n=3$), 73.61 at $\lambda = 300$ ($n=10$) and 73.85 at $\lambda = 1000$ ($n=3$) — the last two sit inside the run-to-run band, so we read the response as saturating rather than continuing to fall. This is the interior optimum that Remark 2 predicts, and it is the reason we treat the large- λ end as a real failure mode rather than as a harmless over-regularization. The seed budgets differ across these points, so we read the shape rather than the individual gaps.

k is a memory–accuracy dial, not a knob to tune. Accuracies at $k \in \{16, 64, 128\}$ differ by less than one point (78.14, 77.87, 77.66) and no paired difference is significant ($p \geq 0.2768$, the $k=128$ comparison; Table 3), while the importance state grows linearly in k — 47.8 MiB at $k=16$ against 182.8 MiB at $k=64$ on the accounting convention of Table 4 (protected directions plus reference snapshot). We report $k=64$ because it was the pre-registered main configuration, not because it scores highest — it does not.

Fisher weighting versus equal weighting is not settled by our runs. Replacing the eigenvalue weights with equal weights per protected direction changes the picture with λ : at $\lambda = 3$ the Fisher-weighted variant is ahead by +3.58 points ($p = 0.019$) over three common seeds, while at $\lambda = 10$ the two are indistinguishable (-0.03 , $p = 0.985$, on the same three seeds; note this is the Fisher-weighted versus equal-weight contrast, not the $\lambda=10$ row of Table 3, which compares that setting against the main configuration over five). With three seeds we claim no ordering in either direction; the comparison would need the main grid’s seed budget before it could carry one, and we flag it here as an open question rather than leaving it out.

5.4. Limitations and discussion

Reproducibility floor. We measured the run-to-run spread directly rather than describing it qualitatively. Two runs on CIFAR-100 share the same seed (0) and the same configuration field by field — their `config.json` files agree on all 21 fields except the `tag`, and the one other field that differs (`folora_lambda`) is not read by the EWC method at all, which we confirmed in the source — so any difference between them is non-determinism rather than a hyperparameter effect. The gap is 77.26 versus 75.97: **1.29 points at the same seed and the same configuration**. That number is not an outlier: the 10-seed spread of the same configuration is ± 1.08 , so it sits inside the distribution. The consequence governs how every comparison in this paper must be read: *the effects we measure on CIFAR-100 are smaller than this single-run noise band*, so a single seed, or any subset of seeds chosen after the fact, cannot support a claim in either direction. This is why we report paired t -tests over seeds throughout and why we describe

any difference inside the band as parity rather than as a win — including differences that happen to favor us.

Where the variance comes from. On CIFAR-100 the accuracy of FOLoRA is consistent in mean but noisier across seeds than the baselines. We attribute this to two sources. First, the importance kernel is estimated from a finite per-sample budget (up to 300 gradients per task), and its top- k approximation is sensitive to the particular samples that land in the tails of the estimate. Second, under a frozen backbone, a fixed-rank ($r=16$) adapter and no replay, the class ordering exerts a strong effect on the final subspace, so individual seeds drift apart. We further found that the epoch budget interacts with the method family: adapter-based Fisher methods degrade under longer training because over-fitting drives the per-sample gradients toward zero and thereby degenerates the estimated Fisher, whereas prompt-based methods benefit from longer training of their prompt keys. This motivates the per-family epoch budgets in Sec. 5.1. A natural direction for future work is a stochastic (Monte-Carlo) Fisher estimate and class-order averaging to tighten the variance.

What we claim, in two layers. Because the CIFAR effect sizes sit inside the noise band above, we separate the claims into one layer the evidence supports unconditionally and one it supports only with an explicit caveat.

The first layer is about *what the method does*, and does not depend on beating anyone. (i) Removing the Fisher-weighted penalty and changing nothing else returns the method to the level of unprotected sequential fine-tuning, so the entire effect of FOLoRA is carried by the regularization term it adds rather than by the shared adapter or the training recipe (Sec. 5.3). (ii)

The protected rank is not a sensitive knob: accuracies at $k \in \{16, 64, 128\}$ differ by less than one point and none of the differences is significant, while the importance state grows linearly in k — which makes k a memory–accuracy dial rather than a hyperparameter to tune. (iii) On CIFAR-100 FOLoRA reaches accuracy *on par with the strongest regularized baseline* rather than above it; we state parity as parity. Against EWC-LoRA, the comparison flips sign depending on which of the tuned EWC operating points is taken as the baseline, so we report all of them in Table 1 and let the reader see the flip rather than selecting the point that reads best.

The second layer is where FOLoRA is *ahead*, and it holds against the parameter-efficient family rather than against full-parameter regularization: against L2P, CODA-Prompt, InfLoRA and O-LoRA, the paired differences are significant on both benchmarks and seed-wise all of one sign. That claim is the one we treat as the paper’s positive result; it is also the one where the baseline’s seed budget is smallest, so Sec. 5.1 records the seed sets explicitly for each family instead of implying a single shared one.

Cost: trainable parameters are traded for importance state. FOLoRA’s efficiency claim is narrower than “memory-efficient”, and we quantify the trade-off rather than leaving it implicit. Its *trainable* parameter count is constant in the number of tasks — the same adapter is reused and re-projected — and the low-rank update merges back into the frozen backbone, so inference costs nothing extra. Against that, the importance model is a per-layer rank- k subspace rather than EWC’s per-parameter diagonal, and it is larger. Table 4 states both sides with a single accounting convention: the importance column counts *everything a method must keep resident to compute its penalty*, i.e. the

Table 4: Resource accounting. Trainable parameters are counted at the first and at the last task ($T=1$ and $T=20$); “importance” is the state a method must keep resident to compute its penalty, counted on one convention for all rows — the diagonal Fisher *plus* the parameter anchor for EWC-LoRA, and the accumulated protected directions *plus* the reference snapshot for FOLoRA (on the same byte counts, EWC-LoRA’s Fisher alone is 2.8 MiB, and FOLoRA’s directions alone are 45.0 MiB at $k=16$ and 180.0 MiB at $k=64$). Inference-token figures are prompt tokens appended to the input; LoRA-based methods merge their update into the backbone and add none.

Method	Trainable params		Grows	Importance state
	$T=1$	$T=20$		
Seq-LoRA	741,125	814,180	no	—
EWC-LoRA	741,125	814,180	no	5.6 MiB
O-LoRA	741,125	14,822,500	yes	—
InfLoRA	741,125	14,822,500	yes	—
L2P	115,205	188,260	yes	—
CODA-Prompt	99,845	172,900	yes	—
FOLoRA ($k=16$)	741,125	814,180	no	47.8 MiB
FOLoRA ($k=64$)	741,125	814,180	no	182.8 MiB

Fisher plus the parameter anchor for EWC, and the accumulated directions plus the reference snapshot for FOLoRA. On that convention FOLoRA at $k=64$ is $32\times$ EWC’s importance state; at $k=16$ it is $8.5\times$. This is the price of representing direction correlations, which a diagonal preconditioner cannot do at any rank, and it is why we report k as a dial rather than claiming a memory advantage.

Scope. Three limitations bound the generality of everything above. First, the evidence covers two benchmarks and a single backbone (ViT-B/16); we make no claim about whether the curvature-weighted subspace transfers to convolutional backbones or to domain-incremental shifts, where the class-incremental protocol we evaluate under does not directly apply. Second, the baselines are reproduced under one shared protocol and one shared backbone rather than at their published settings, which is what makes the comparison internally fair but also means the numbers are not the numbers reported in the original papers; where a baseline’s own configuration is materially different — the prompt methods’ pool size and prompt length, and the number of training epochs for the InfLoRA comparison — we state both configurations, so that the comparison cannot be read as a deliberately weakened baseline. Third, the subspace baselines fall below the frozen-feature floor on CIFAR-100 — InfLoRA at 70.14 and O-LoRA at 64.79, against a floor of 70.31 — but not on *both* benchmarks: on ImageNet-R O-LoRA remains below the floor (44.79 against 49.71) whereas InfLoRA lands marginally *above* it (50.00, i.e. +0.29). We report every variant we ran rather than only the weakest. For InfLoRA that is both the five- and the twenty-epoch configuration (70.14/70.01 on CIFAR-100, 50.00/49.78 on ImageNet-R). For O-LoRA it is both the initialization-only row of Table 1 and the training-time orthogonalized objective that the original paper specifies — the formulation augments the loss with $\lambda_1 \sum_{i < t} \|A_t^\top A_i\|_F^2$, and the row we tabulate implements the orthogonal initialization without it. We ran that objective at two constraint strengths, $\lambda_1=0.1$ and $\lambda_1=1.0$; they reach 68.44 ± 0.50 and 68.44 ± 1.07 on CIFAR-100 and 48.14 and 48.48 on ImageNet-R ($n=3$ each): better than the

tabulated row, but still below the floor on both benchmarks. Against these stronger configurations FOLoRA’s advantage narrows from +13.07 to +9.93 on CIFAR-100, at $p = 0.005$ for $\lambda_1=0.1$ and $p = 0.012$ for $\lambda_1=1.0$; we quote the weaker of the two, which remains significant. adapter input space does not control the norm of the merged update, which is what a task-ID-free evaluation reads. We treat that as a finding about task-ID-free merging at $T=20$ rather than as evidence about the methods themselves, and it is why we do not present FOLoRA’s margin over that family as a merit comparison.

The bound’s slack. Finally, Sec. 4 reports that the forgetting bound is not tight at the operating point we probe: its remainder and its stationarity residual are the same order as its leading term, and the finite-sample Fisher estimate is not a calibrated surrogate for the measured curvature. The derivation is therefore used as a statement of *which* quantity to penalize, and the empirical support for the method is the ablation above rather than the bound. We prefer to state this explicitly, since the alternative — presenting a loose bound as if it were predictive — is the failure mode that a reader who re-derives Eq. (15) would find anyway.

6. Conclusion

We propose FOLoRA, a replay-free method for continual learning that mitigates catastrophic forgetting in low-rank adaptation while keeping the number of trainable parameters fixed as the number of tasks grows. FOLoRA maintains an accumulated second-order importance kernel for each adapted layer, forms an importance-weighted protected subspace from its top eigendirections, and penalizes alignment of the adapter’s displacement with those

directions in proportion to their importance. We derive a forgetting upper bound that decomposes into Fisher-weighted subspace overlaps and showed that FOLoRA’s regularizer minimizes its dominant term, thereby providing a principled stability–plasticity trade-off and containing equal-weight orthogonalization as a special case. We also measure the slack of that bound instead of assuming it away: at the operating point we probe the remainder and the stationarity residual are the same order as the quadratic term, so the bound is presented as a design principle for what to penalize rather than as a tight or numerically predictive characterization of forgetting. The empirical case for the regularizer rests on the $\lambda=0$ ablation, which removes the Fisher-weighted penalty while changing nothing else and returns the method to the level of unprotected sequential fine-tuning. Experiments on standard class-incremental benchmarks show that FOLoRA reaches an average accuracy on par with the strongest regularized baseline while using no replay buffer and no growth in the parameter count, and we quantify the importance-model memory it pays for representing that model as a rank- k subspace rather than EWC’s diagonal one — $32\times$ EWC’s importance state at $k=64$ and $8.5\times$ at $k=16$, which is the trade-off that makes k a memory–accuracy dial rather than a free parameter. We also report the run-to-run reproducibility floor we measured (1.29 points at a fixed seed and a fixed configuration), and we state the resulting conclusions at the strength the data supports: parity with the strongest regularized baseline, rather than an improvement over it, and a significant advantage over the parameter-efficient prompt and subspace baselines.

A natural direction for future work is to extend FOLoRA to task-agnostic

settings with distribution shift (domain-incremental learning) and to language models, where the low-rank subspace geometry of different task families can be exploited more explicitly.

CRediT authorship contribution statement

Jiang Tao: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study (CIFAR-100 and ImageNet-R) are publicly available standard continual-learning benchmarks. The source code implementing the proposed method, all baselines, the evaluation protocol and the scripts that reproduce every table and figure in this paper are openly available at <https://github.com/ml-universe/FOLoRA>.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) in order to draft, restructure and edit the manuscript text. After using this

tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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References

- [1] R. M. French, Catastrophic forgetting in connectionist networks, Trends in Cognitive Sciences 3 (4) (1999) 128–135.
- [2] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, icarl: Incremental classifier and representation learning, in: CVPR, 2017.
- [3] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, et al., Overcoming catastrophic forgetting in neural networks, Proceedings of the National Academy of Sciences 114 (13) (2017) 3521–3526.
- [4] F. Zenke, B. Poole, S. Ganguli, Continual learning through synaptic intelligence, in: ICML, 2017.
- [5] A. A. Rusu, N. C. Rabinowitz, G. Desjardins, et al., Progressive neural networks, arXiv preprint arXiv:1606.04671 (2016).
URL <https://arxiv.org/abs/1606.04671>
- [6] Z. Wang, Z. Zhang, C.-Y. Lee, et al., Learning to prompt for continual learning, in: CVPR, 2022.

- [7] J. S. Smith, L. Karlinsky, V. Gutta, et al., Coda-prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning, in: CVPR, 2023.
- [8] X. Wang, T. Chen, Q. Ge, et al., Orthogonal subspace learning for language model continual learning, in: Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 10658–10671. doi:10.18653/v1/2023.findings-emnlp.715.
- [9] D.-W. Zhou, H.-L. Sun, J. Ning, H.-J. Ye, D.-C. Zhan, Continual learning with pre-trained models: A survey, in: Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI), 2024, pp. 8363–8371. doi:10.24963/ijcai.2024/924.
- [10] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting class-incremental learning with pre-trained models: Generalizability and adaptivity are all you need, International Journal of Computer Vision 133 (3) (2025) 1012–1032. doi:10.1007/s11263-024-02218-0.
- [11] E. N. Coleman, L. Quarantiello, Z. Liu, Q. Yang, S. Mukherjee, J. Hurtado, V. Lomonaco, Parameter-efficient continual fine-tuning: A survey, Neurocomputing 701 (2026) 134502. doi:10.1016/j.neucom.2026.134502.
- [12] R. Razavi-Far, E. Hallaji, A. Fathalizadeh, M. Wu, M. Rostami, Towards self-adaptive learning: A comprehensive survey on continual learning under harsh conditions, Neurocomputing 698 (2026) 134136. doi:10.1016/j.neucom.2026.134136.

- [13] J. Leo, J. Kalita, Survey of continuous deep learning methods and techniques used for incremental learning, *Neurocomputing* 582 (2024) 127545. doi:10.1016/j.neucom.2024.127545.
- [14] E. J. Hu, Y. Shen, P. Wallis, et al., Lora: Low-rank adaptation of large language models, in: *International Conference on Learning Representations (ICLR)*, 2022.
- [15] Y. Wu, H. Piao, L.-K. Huang, R. Wang, W. Li, H. Pfister, D. Meng, K. Ma, Y. Wei, SD-LoRA: Scalable decoupled low-rank adaptation for class incremental learning, in: *International Conference on Learning Representations (ICLR)*, 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/92f43b1d33fae4aa1958f75317f0cec1-Abstract-Conference.html
- [16] J. He, Z. Duan, F. Zhu, CL-LoRA: Continual low-rank adaptation for rehearsal-free class-incremental learning, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 30534–30544. doi:10.1109/CVPR52734.2025.02843.
- [17] H. Zhu, Y. Zhang, J. Dong, P. Koniusz, BiLoRA: Almost-orthogonal parameter spaces for continual learning, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 25613–25622. doi:10.1109/CVPR52734.2025.02385.
- [18] X. Wei, G. Li, R. Marculescu, Online-LoRA: Task-free online continual

- learning via low rank adaptation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025, pp. 6634–6645. doi:10.1109/WACV61041.2025.00646.
- [19] Y.-S. Liang, W.-J. Li, Inflora: Interference-free low-rank adaptation for continual learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 23638–23647. doi:10.1109/CVPR52733.2024.02231.
- [20] D.-W. Zhou, H.-L. Sun, H.-J. Ye, D.-C. Zhan, Expandable subspace ensemble for pre-trained model-based class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, pp. 23554–23564. doi:10.1109/CVPR52733.2024.02223.
- [21] X. Wang, Z. Zhuang, Y. Zhang, PLAN: Proactive low-rank allocation for continual learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025, pp. 2909–2918. doi:10.1109/ICCV51701.2025.00279.
- [22] J. Huang, L. Xu, Y. Liu, Z. Li, Reducing catastrophic forgetting via gradient-based task organization in class-incremental learning, *Neurocomputing* 706 (2026) 134964. doi:10.1016/j.neucom.2026.134964.
- [23] R. Aljundi, F. Babiloni, M. Elhoseiny, et al., Memory aware synapses: Learning what (not) to forget, in: ECCV, 2018.
- [24] A. Mallya, S. Lazebnik, Packnet: Adding multiple tasks to a single network by iterative pruning, in: Proceedings of the IEEE/CVF Con-

- ference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 7765–7773. doi:10.1109/CVPR.2018.00810.
- [25] A. Cossu, F. Giannini, G. Ziffer, A. Bernardo, A. Gepperth, E. Della Valle, B. Hammer, D. Bacciu, A practical guide to streaming continual learning, *Neurocomputing* 674 (2026) 132951. doi:10.1016/j.neucom.2026.132951.
 - [26] K. Li, C. Ding, J. Wan, S. Yu, Enhance the old representations’ adaptability dynamically for exemplar-free continual learning, *Neurocomputing* 639 (2025) 130286. doi:10.1016/j.neucom.2025.130286.
 - [27] Q. Ferdinand, B. Clement, P. Papadakis, Q. Oliveau, G. Le Chenadec, Feature expansion and enhanced compression for class incremental learning, *Neurocomputing* 634 (2025) 129807. doi:10.1016/j.neucom.2025.129807.
 - [28] M. Liu, F. Ye, Learning diverse and adaptive representations for continual learning, *Neurocomputing* 664 (2026) 132112. doi:10.1016/j.neucom.2025.132112.
 - [29] A. Carta, A. Cossu, V. Lomonaco, D. Bacciu, J. van de Weijer, Projected latent distillation for data-agnostic consolidation in distributed continual learning, *Neurocomputing* 598 (2024) 127935. doi:10.1016/j.neucom.2024.127935.
 - [30] Y. Yang, J. Zhou, X. Ding, T. Huai, S. Liu, Q. Chen, Y. Xie, L. He, Recent advances of foundation language models-based continual learn-

- ing: A survey, *ACM Computing Surveys* 57 (5) (2025) 1–38. doi:10.1145/3705725.
- [31] Z. Wang, Z. Zhang, S. Ebrahimi, R. Sun, C.-Y. Lee, S. Ren, G. Su, V. Perot, J. Dy, T. Pfister, Dualprompt: Complementary prompting for rehearsal-free continual learning, in: *ECCV*, 2022.
 - [32] W. Fang, H. Che, F. Ren, J. Jiang, Q. Lao, Normality-preserving continual industrial anomaly detection via orthogonal LoRA banks, *Neurocomputing* 706 (2026) 135021. doi:10.1016/j.neucom.2026.135021.
 - [33] Q. Chen, J. Huang, W. Wen, Q. Li, R. Zhang, J. Qin, CAT: Continual adapter tuning for aspect sentiment classification, *Neurocomputing* 580 (2024) 127423. doi:10.1016/j.neucom.2024.127423.
 - [34] M. Li, P. Ye, L. Zhang, B. Bai, T. Chen, Adapter-X: A general parameter-efficient fine-tuning framework for 2d and 3d vision, *Neurocomputing* 677 (2026) 133113. doi:10.1016/j.neucom.2026.133113.
 - [35] P. Wang, J. Wang, Z. Zhong, W. Bao, Y. Zhang, J. Chen, Enhancing long-term memory in federated class continual learning with lightweight adapters, *Neurocomputing* 656 (2025) 131459. doi:10.1016/j.neucom.2025.131459.
 - [36] D. Luo, K. Zheng, C. Wu, X. Wang, J. Wang, ERAT-DLoRA: Parameter-efficient tuning with enhanced range adaptation in time and depth aware dynamic LoRA, *Neurocomputing* 614 (2025) 128778. doi:10.1016/j.neucom.2024.128778.

- [37] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, L. Zhang, D.-C. Zhan, Dual consolidation for pre-trained model-based domain-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025, pp. 20547–20557. doi:10.1109/CVPR52734.2025.01913.
- [38] L. Wang, J. Xie, X. Zhang, M. Huang, H. Su, J. Zhu, Hierarchical decomposition of prompt-based continual learning: Rethinking obscured sub-optimality, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 36, 2023, pp. 69054–69076. doi:10.52202/075280-3022.
- [39] L. Pan, Z. Lu, Y. Zheng, C. Yan, H. Wen, L. Tao, C. Xu, Task-adaptive continual learning of vision language models via prototype routing and prompt, Neurocomputing 672 (2026) 132740. doi:10.1016/j.neucom.2026.132740.
- [40] X. Xi, G. Cao, W. Cao, Y. Liu, Y. Li, H. Wang, H. Ren, Potential knowledge extraction network for class-incremental learning, Neurocomputing 616 (2025) 128923. doi:10.1016/j.neucom.2024.128923.
- [41] C. Rebillard, J. Hurtado, A. Krutsylo, L. Passaro, V. Lomonaco, Continually learn to map visual concepts to language models in resource-constrained environments, Neurocomputing 652 (2025) 131013. doi:10.1016/j.neucom.2025.131013.
- [42] H. Zhao, R. Liu, Y. Liu, Z. Zou, Y. Chen, Decoupled and distilled: Task-adaptive LoRA-teachers with ensemble knowledge transfer for few-shot

- class-incremental learning, *Neurocomputing* 708 (2027) 135200. doi:
10.1016/j.neucom.2026.135200.
- [43] M. Farajtabar, N. Azizan, A. Mott, A. Li, Orthogonal gradient descent for continual learning, in: *AISTATS*, 2020.
- [44] G. Saha, I. Garg, K. Roy, Gradient projection memory for continual learning, in: *ICLR*, 2021.
- [45] S. Wang, X. Li, J. Sun, Z. Xu, Training networks in null space of feature covariance for continual learning, in: *CVPR*, 2021, pp. 184–193. doi:
10.1109/CVPR46437.2021.00025.
- [46] M. A. Bennani, M. Sugiyama, Generalisation guarantees for continual learning with orthogonal gradient descent, *arXiv preprint arXiv:1906.06195* (2019).
URL <https://arxiv.org/abs/1906.06195>
- [47] Y. Lu, S. Zhang, D. Cheng, Y. Xing, N. Wang, P. Wang, Y. Zhang, Visual prompt tuning in null space for continual learning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2024, arXiv:2406.05658.
URL <https://arxiv.org/abs/2406.05658>
- [48] E. Yang, L. Shen, Z. Wang, S. Liu, G. Guo, X. Wang, D. Tao, Revisiting flatness-aware optimization in continual learning with orthogonal gradient projection, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 47 (5) (2025) 3895–3907. doi:10.1109/TPAMI.2025.3539019.

- [49] C. Liu, Q. Liu, R. Li, X. Zhou, M. S. Kiran, X. Wang, A study on generalization of random weight network with flat loss, *Neurocomputing* 657 (2025) 131650. doi:10.1016/j.neucom.2025.131650.
- [50] X. Wang, T. Liu, M. Zhou, Y. Zhang, Z. Yuan, K. Liu, P. Yang, Adapting to dissimilar tasks for continual learning via gradient norm regularisation, *Neurocomputing* 695 (2026) 134015. doi:10.1016/j.neucom.2026.134015.
- [51] C. V. Nguyen, A. Achille, M. Lam, T. Hassner, V. Mahadevan, S. Soatto, Toward understanding catastrophic forgetting in continual learning, *arXiv preprint arXiv:1908.01091* (2019).
URL <https://arxiv.org/abs/1908.01091>
- [52] I. Evron, E. Moroshko, R. Ward, N. Srebro, D. Soudry, How catastrophic can catastrophic forgetting be in linear networks?, in: *Advances in Neural Information Processing Systems*, 2022.
- [53] R. Karakida, S. Akaho, Learning curves for continual learning in neural networks: Self-knowledge transfer and forgetting, in: *ICLR*, 2022.
- [54] T. Doan, M. A. Bennani, B. Mazouze, G. Rabusseau, P. Alquier, A theoretical analysis of catastrophic forgetting through the ntk overlap matrix, in: *AISTATS*, 2021.
- [55] J. Galdi, A. Breccia, G. Lanzillotta, T. Hofmann, L. Noci, The importance of being lazy: Scaling limits of continual learning, in: *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Vol.

267 of Proceedings of Machine Learning Research, 2025.

URL <https://proceedings.mlr.press/v267/graldi25a.html>

- [56] G. Zheng, P. Wang, S. Hu, Q. Zheng, L. Shen, Understanding the dynamics of forgetting and generalization in continual learning via the neural tangent kernel, in: International Conference on Learning Representations (ICLR), 2026.

URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/1f8638a87bddd34e090698f0bfaa9c02-Abstract-Conference.html

- [57] R. Shuttleworth, J. Andreas, A. Torralba, P. Sharma, LoRA vs full fine-tuning: An illusion of equivalence, in: Advances in Neural Information Processing Systems (NeurIPS), 2025, pp. 193421–193456. doi:10.52202/085713-5808.

- [58] G. Lai, D.-W. Zhou, X. Yang, H.-J. Ye, The lie of the average: How class incremental learning evaluation deceives you?, arXiv preprint arXiv:2509.22580 (2026).

URL <https://arxiv.org/abs/2509.22580>

- [59] S. Cha, J. Kwak, D. Shim, H. Kim, M. Lee, H. Lee, T. Moon, Towards more diverse evaluation of class incremental learning: Representation learning perspective, in: Proceedings of the Conference on Lifelong Learning Agents (CoLLAs), Vol. 274 of Proceedings of Machine Learning Research, 2025, pp. 86–101.

URL <https://proceedings.mlr.press/v274/cha25a.html>

- [60] G. Lee, Y. Oh, G. Jang, S. Lee, J. Song, S. Cha, Y. Yoo, Continual-
MEGA: A large-scale benchmark for generalizable continual anomaly de-
tection, *Neurocomputing* 700 (2026) 134460. doi:10.1016/j.neucom.
2026.134460.
- [61] J. Pomponi, A. Devoto, S. Scardapane, Class incremental learning with
probability dampening and cascaded gated classifier, *Neurocomputing*
640 (2025) 130295. doi:10.1016/j.neucom.2025.130295.
- [62] W. Zhang, P. Hu, Q. Li, J. Tan, Z. Tan, R. Wang, Boosting confidence
in class-incremental learning via discriminative local feature enhance-
ment, *Neurocomputing* 660 (2026) 131829. doi:10.1016/j.neucom.
2025.131829.
- [63] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den
Hengel, RanPAC: Random projections and pre-trained models for con-
tinual learning, in: *Advances in Neural Information Processing Sys-
tems (NeurIPS)*, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/
075280-0526.
- [64] D. Castellana, A. Carta, CD-IMM: A Bayesian non-parametric classifier
for continual learning, *Neurocomputing* 704 (2026) 134751. doi:10.
1016/j.neucom.2026.134751.
- [65] C. T. Tran, T. T. Nguyen, T. P. Nguyen, N. Thirion-Moreau, Geometry-
controlled convex hull prototype framework for online task-free continual
learning, *Neurocomputing* 701 (2026) 134619. doi:10.1016/j.neucom.
2026.134619.

- [66] J. Martens, R. Grosse, Optimizing neural networks with kronecker-factored approximate curvature, in: Proceedings of the 32nd International Conference on Machine Learning (ICML), Vol. 37 of Proceedings of Machine Learning Research, 2015, pp. 2408–2417.
- [67] F. Kunstner, L. Balles, P. Hennig, Limitations of the empirical fisher approximation for natural gradient descent, in: NeurIPS, 2019.